MILITARY SPECIFICATION SHEET

ELECTRON TUBE, CATHODE RAY

TYPE 7AEP7

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Electrostatic focus and deflection, flat face, single beam, post-accelerator

DIMENSIONS AND PIN CONNECTIONS: See figure 1

ABSOLUTE RATINGS:

Parameter:	Ef	Ec1	Eb1	Eb2	Eb3	ed	Rg	Zd	Ehk	Eb3/Eb2	Alt
Unit:	V	Vdc	Vdc	Vdc	Vdc	v	Meg	Meg	Vdc	Ratio (Note 1)	ft
Maximum:	6.9	0	1,375	4,400	8,800	825	1.5	1.0 (Note 2)	-180	2.0	10,000
Minimum:	5.7	-200	---	1,000	1,000	---	---	---	---	1.0	---
TEST CONDITIONS:	6.3	Adj	Focus	2,000	4,000	---	---	---	---	2.0	---

GENERAL:

Qualification - Not required

(B) denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	Quality conformance inspection, part 1						
5201	Voltage breakdown	-		---	---	---	---
5201	Voltage breakdown (electrostatic type)	-		---	---	---	---
5206	Gas "cross"	4	Ib3 - 25 µAdc	---	---	---	---
5106	Bulb, screen, and faceplate quality	-		---	---	---	---
5223	Modulation	-	Ib3 = 25 µAdc	ΔEc1	---	37	Vdc
5231	Spot position (electrostatic deflection)	-		---	---	---	---
5231	Spot displacement (leakage)	-		Displ	---	10	mm
5241	Grid cutoff voltage	-		Eco	-45	-75	Vdc
5251	Grid No. 1 leakage current	-		---	---	---	---
5251	Anode No. 1 leakage current	-		---	---	---	---
5251	Anode No. 2 leakage current	-		---	---	---	---
	Quality conformance inspection, part 2						
1301	Heater current	-		If	540	660	mA
5201	Electrode current (anode No. 1)	-	Ec1 - 0	Ib1	-15	+10	µAdc
5101	Base alinement (electrostatic types)	-	+1D2; pin No. 5	---	---	---	---
5101	Side terminal alinement (electrostatic types)	-	+1D2	---	---	---	---
5101	Side terminal and base alinement	-	Pin No. 5	---	---	---	---
5101	Neck and base alinement (electrostatic types)	-		---	---	---	---
5101	Angle between traces	-		---	89	91	Degrees
5216	Cathode illumination	-		---	---	$1.5X10^{-4}$	ftL
(B)5216	Stray light emission (conventional types)	-	Eb2 = 4,400 Vdc; Eb3 = 8,800 Vdc	---	---	---	---
5221	Screens	-		---	---	---	---
5216	Line width "A" (electrostatic deflection)	-	Ib3 = 25 µAdc	Width	---	0.55	mm
5226	Line width "B" (electrostatic deflection)	-	Ib3 = 25 µAdc	Width	---	0.80	mm
5246	Focusing voltage at cutoff	-		Eb1	---	620	Vdc
5246	Focusing voltage	-		Eb1	380	---	Vdc
5248	Deflection factor	-	1D2	DF	81	101	Vdc/in.
5248	Deflection factor	-	3D4	DF	67	83	Vdc/in.

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	Quality conformance inspection, part 2 - Continued						
5251	Heater - cathode leakage current	-		---	---	---	---
5201	Electrode current (cathode)	3	Ib3 = 25 µAdc	Ik	---	500	µAdc
1101	Secureness of base, cap, or insert	-		---	---	---	---
(8) 1111	Base pin solder depth	-		---	---	---	---
(8) 1105	Permanence of marking	-		---	---	---	---
	Quality conformance inspection, part 3						
---	Life test	-	Group C; Ib3 = 25 µAdc; Eb2 = 4,400 Vdc; t = 500 hours	---	---	---	---
---	Life - test end points:						
5223	Modulation	-	Ib3 = 18.75 µAdc	ΔEc1	---	37	Vdc
5226	Line width "A"	-	Ib3 = 18.75 µAdc	Width	---	0.55	mm
5226	Line width "B"	-	Ib3 = 18.75 µAdc	Width	---	0.80	mm

NOTES:

1. This tube is designed for optimum performance when operating at an Eb3/Eb2 ratio of 2.0. Operation at other ratios of Eb3/Eb2 may result in changes in deflection uniformity.
2. It is recommended that the deflecting electrode circuit resistances be approximately equal. Higher resistances values up to 5.0 megohms may be used for low beam current operation.
3. Accelerator power input (average) should be limited to 6 watts.
4. This test to be performed at the conclusion of the holding period.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Air Force - 11, 17, 80
DSA - ES

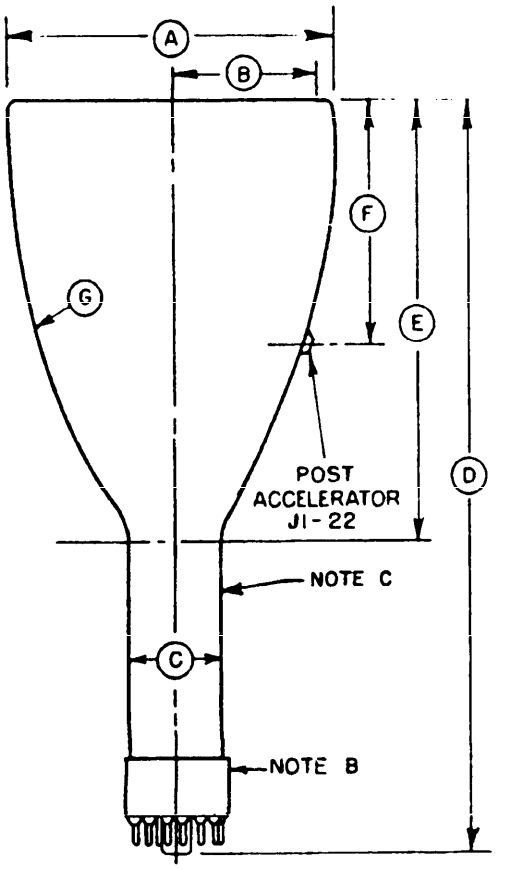
User activities:
Army -
Navy - AS, OS, MC, CG, SH

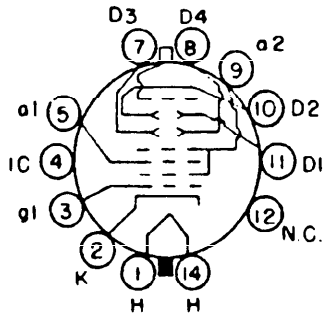
Preparing activity: Navy - EC

Agent: DSA - ES

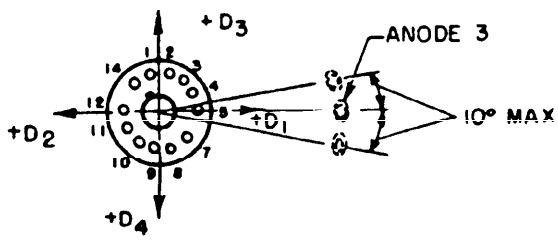
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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 2		
A	6.906 (175.41)	7.094 (180.19)
B	See note a	
C	1.937 (49.20)	2.063 (52.40)
D	15.25 (387.35)	15.75 (400.05)
F	4.75 (120.65)	5.25 (133.35)
Reference dimensions (see note d)		
E	9.00 (228.60) Nom.	
G	13.00 (330.20) Rad.	



NOTES:
a. The minimum useful screen radius shall be not less than 3.063 (77.80 mm) inches.
b. The base shall be a medium shell 12 pin diheptal base (B 12-37).
c. The bulb shall be a J56Y1 type.
d. These dimensions are for information only and are not required for inspection purposes.

FIGURE 1. Outline drawing of electron tube type 7AEP7.

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NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D. C. 20360

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